

Mitochondria and cellular respiration

Mitochondria are the organelles where cellular respiration takes place. Glucose is broken down in glycolysis in the cytoplasm, producing pyruvate and a small amount of ATP.

The Krebs cycle runs in the mitochondrial matrix. Each turn releases carbon dioxide and loads the electron carriers NADH and FADH₂.

The electron transport chain

On the inner mitochondrial membrane, the electron transport chain pumps protons into the intermembrane space. ATP synthase lets them flow back and uses that gradient to make most of the cell's ATP. Oxygen is the final electron acceptor and combines with protons to form water.

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